PRACTICAL 8. Ethics and Bias Analysis

Aim:- Select any publicly available dataset (for example, Iris, Adult Income, or a classification dataset of your choice) and analyze it for possible bias and class Imbalance.

Code:-

```
import pandas as pd
from sklearn.datasets import load_iris
iris=load_iris()
data=pd.DataFrame(
    iris.data,
    columns=iris.feature_names
)

data['Class']=iris.target
print("First 5 record")
print(data.head())
print("\nClass Distribution")
print(data['Class'].value_counts())

print("Missing Values:")
print(data.isnull().sum())
```

Output:-

```
First 5 record
   sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)  \
0                5.1               3.5                1.4               0.2
1                4.9               3.0                1.4               0.2
2                4.7               3.2                1.3               0.2
3                4.6               3.1                1.5               0.2
4                5.0               3.6                1.4               0.2

   Class
0      0
1      0
2      0
3      0
4      0
```

Class Distribution
Class
0 50
1 50
2 50

Name: count, dtype: int64
Missing Values:
sepal length (cm) 0
sepal width (cm) 0
petal length (cm) 0
petal width (cm) 0
Class 0
dtype: int64